
purehw

Name

purehw, purehw-list, purehw-setattr — displays information about and controls visual identification of FlashArray hardware components

Synopsis

```
purehw list [ --csv | --nvp ] [--notitle] [--page] [--raw] [--spec] [--type COMPONENT-TYPE] [COMPONENT...]
```

```
purehw setattr [ --identify {off | on} | --index INDEX ] COMPONENT...
```

Arguments

COMPONENT

Hardware component whose information is to be displayed or whose attribute is to be set to the specified value.

Options

-h | --help

Can be used with any command or subcommand to display a brief syntax description.

--identify

Turns a visual identifier for the component on or off. Valid for controllers, NVRAM bays, storage bays, and storage shelves. Used with the **purehw setattr** command to set front panel LED identifiers on or off.

--index

In multi-shelf arrays, sets the top-level shelf component to a unique number. Once set, the number becomes part of the name of each component in the shelf.

--spec

Displays hardware component model names, part numbers, and serial numbers.

--type

Type of component for which information is to be displayed. When this option is specified, information is displayed for all components of the specified type. Valid values for `--type` are: `ch` (chassis), `ct` (controller), `bay` (flash module bay), `eth` (Ethernet port), `fan` (fan), `fc` (Fibre Channel port), `ib` (InfiniBand port), `iom` (I/O module), `nvb` (NVRAM module bay), `pwr` (power supply), `sas` (SAS port), `tmp` (temperature sensor), and `sh` (storage shelf).

Supported `--type` values vary by FlashArray model.

Options that control display format:

--csv

Lists information in comma-separated value (CSV) format. The `--csv` output can be used for scripting purposes and imported into spreadsheet programs.

--notitle

Lists information without column titles.

--nvp

Lists information in name-value pair (NVP) format, in the form `ITEMNAME=VALUE`. Argument names and information items are displayed flush left. The `--nvp` output is designed both for convenient viewing of what might otherwise be wide listings, and for parsing individual items for scripting purposes.

--page

Turns on interactive paging.

--raw

Displays the unformatted version of column titles and data. For example, in the `purearray monitor` output, the unformatted version of column title `us/op (read)` is `usec_per_read_op`. The `--raw` output is used to sort and filter list results.

Description

Most FlashArray hardware components report their operational status. The **purehw list** command displays information about the FlashArray hardware components. The **purehw setattr** command controls the visual identification of certain hardware components, and sets numerical values of top-level shelf components.

FlashArray Hardware Components

Hardware components and their naming vary by FlashArray series. To see the hardware technical specifications for each FlashArray model, refer to the Products page at <https://www.purestorage.com>.

Hardware Components in FlashArray//X and FlashArray//M

The //X and //M chassis, controller and storage shelf names have the form `XXm`. The names of other components have the form `XXm.YYn` or `XXm.YYYn`, where:

`XX`

Denotes the type of component:

- `CH` - //X or //M chassis.
 - `CT` - Controller.
 - `SH` - Storage shelf.
-

`m`

Identifies the specific controller or storage shelf:

- For an //X or //M chassis, `m` has a value of 0. For example, `CH0`.
- For controllers, `m` has a value of 0 or 1. For example, `CT0`, `CT1`.
- For storage shelves, `m` represents the shelf number, starting at 0. For example, `SH0`, `SH1`.

The assigned number can be changed on the shelf front panel or by running the `purehw setattr --id` command.

`YY or YYY`

Denotes the type of component. For example, `FAN` for cooling device, `FC` for Fibre Channel port.

`n`

Identifies the specific component by its index (its relative position within the //X or //M chassis, controller, or storage shelf), starting at 0.

The following table lists //X and //M hardware components that report status, grouped by their location on the array. The Identify Light column shows which components have an LED light on the physical component that can be turned on and off.

Chassis (CH0)

Component Name	Identify Light	Component Type
CH0	Yes	Chassis
CH0.BAYn	Yes	Storage bay
CH0.NVBn	Yes	NVRAM bay
CH0.PWRn	--	Power module

Controller (CTm)

Component Name	Identify Light	Component Type
CTm	Yes	Controller
CTm.ETHn	--	Ethernet port
CTm.FANn	--	Cooling fan
CTm.FCn	--	Fibre Channel port
CTm.IBn	--	InfiniBand port (included only with certain upgrade kits)
CTm.SASn	--	SAS port
CTm.TMPn	--	Temperature sensor

Storage Shelf (SHm)

Component Name	Identify Light	Component Type
SHm	Yes	Storage shelf
SHm.BAYn	Yes	Storage bay
SHm.FANn	--	Cooling fan
SHm.IOMn	--	I/O module
SHm.PWRn	--	Power module
SHm.SASn	--	SAS port

Component Name	Identify Light	Component Type
SHm.TMPn	--	Temperature sensor

FlashArray//X and FlashArray//M Chassis Components (CH0)

In //X and //M, the chassis (**CH0**) contains the controllers and the bays that host the Flash Modules and NVRAM modules. The controller component names use **CT0**. The component names for the bays use **CH0**. The chassis status is generated from a combination of software conditions and the status of its components.

The components in an //X or //M chassis include:

- **Storage Bay (CH0.BAYn):**

Each storage shelf contains a row of hot-swappable storage bays, which contain flash modules used for the array's data. Storage bays are indexed from left to right starting at 0.

- **NVRAM Bay (CH0.NVBn):**

The chassis contains multiple NVRAM bays, which are hot-swappable and house NVRAM modules (temporary holding areas for host-written data). Bays are indexed from left to right, facing the front of the shelf.

- **Power Supply (CH0.PWRn):**

The internal shelf contains redundant power supplies that report status. Power supply replacement is a service technician operation.

FlashArray//X or FlashArray//M Controller Components (CTm)

Purity//FA reports an overall controller (**CTm**) status. The controller status is generated from a combination of software conditions and the status of its components. For example, the status **ok** is reported for properly functioning controllers. Controllers contain 2- or 4-port Fibre Channel interface cards, 2-port Ethernet I/O interface cards, or a mix of both types.

The components in an //X or //M controller include:

- **Ethernet Port (CTm.ETHn):**

Ethernet ports report status and link speed. Each controller contains at least 1x1GbE port for management (including one used for remote array administration) and 2x10GbE ports for replication.

- **Fan (CTm.FANn):**

Each controller contains multiple cooling fans that report status. Fan replacement is a service technician operation.

- **Fibre Channel Port (CTm.FCn):**

Fibre Channel ports report status and link speed. Most Fibre Channel-based controllers contain redundant 2- or 4-port Fibre Channel host bus adapters. Some controllers support a single Fibre Channel adapter. To see the Fibre Channel port details for each FlashArray model, refer to the Products page at <https://support.purestorage.com>.

- **InfiniBand Port (CTm.IBn):**

InfiniBand ports are shipped only in upgrade kits for FA-4xx-to-FlashArray//X or //M upgrades. FlashArray//X and //M use InfiniBand ports only for controller upgrades from FA-4xx systems. InfiniBand ports report status and communication speed.

- **SAS Ports (CTm.SAS):**

Each controller contains multiple dual-port SAS interface cards that communicate with flash modules. SAS ports report status and communication speed.

- **Temperature Sensor (CTm.TMPn):**

Each controller chassis contains multiple temperature sensors that report their status and the ambient temperature at various points within the chassis. In steady-state operation, Purity//FA generates alerts when a temperature sensor reports a value outside of normal operating range.

Expansion Shelf Components (SHm)

Purity//FA reports an overall storage shelf (**SHm**) status generated from a combination of software conditions and the status of its components.

The optional expansion shelf is available with most //M models.

The components in an expansion shelf include:

- **Storage Bay (SHm.BAYn):**

Each storage shelf contains a row of hot-swappable storage bays, which contain flash modules used for the array's data. Storage bays are indexed from left to right starting at 0.

- **Fan (SHm.FANn):**

Each of a shelf's power and cooling modules (PCMs) reports cooling fan status.

- **IOM (SHm.IOMn):**

Redundant I/O Modules (IOMs) accessed from the rear of the shelf chassis contain the shelf's SAS ports.

- **Power Supply (SHm.PWRn):**

The power supply sub-component within each of a shelf's power and cooling modules (PCMs) reports its operating status. Power supply replacement is a service technician operation.

- **SAS Ports (SHm.SASn):**

Multiple SAS ports handle communication between the array's controllers and the shelf's flash modules.

- **Temperature Sensor (SHm.TMPn):**

Each controller chassis contains multiple temperature sensors that report their status and the ambient temperature at various points within the chassis. In steady-state operation, Purity//FA generates alerts when a temperature sensor reports a value outside of normal operating range.

Hardware Components in the FA-400 Series

In the FA-400 series, the controller and storage shelf names have the form `XXm`. The names of other components have the form `XXm.YYn` or `XXm.YYYn`, where:

`XX`

Denotes the type of chassis. Controllers use `CT`. Storage shelves use `SH`.

`m`

Identifies the specific controller or storage shelf.

- For controllers, `m` has a value of 0 or 1. For example, `CT0`, `CT1`.
- For storage shelves, `m` represents the shelf number, starting at 0. For example, `SH0`, `SH1`.

The assigned number can be changed on the shelf front panel or by running the `purehw setattr --id` command.

`YY or YYY`

Denotes the type of component. For example, `FAN` for cooling device, `FC` for Fibre Channel port.

`n`

Identifies the specific component by its index (its relative position within the array), starting at 0.

The following table lists FA-400 hardware components that report status, grouped by their location on the array. The Identify Light column shows which components have an LED light on the physical component that can be turned on and off.

Controller (CTm)

Component Name	Identify Light	Component Type
CTm	Yes	Controller
CTm.ETHn	--	Ethernet port
CTm.FANn	--	Cooling fan
CTm.FCn	--	Fibre Channel port
CTm.IBn	--	InfiniBand port
CTm.PWRn	--	Power module
CTm.SASn	--	SAS port
CTm.TMPn	--	Temperature sensor

Storage Shelf (SHm)

Component Name	Identify Light	Component Type
SHn	Yes	Storage shelf
SHn.BAYn	Yes	Storage bay
SHn.FANn	--	Cooling fan
SHn.IOMn	--	I/O module
SHn.PWRn	--	Power module
SHn.SASn	--	SAS port
SHn.TMPn	--	Temperature sensor

FA-400 Controller Components (CTm)

Purity//FA reports an overall controller (**CTm**) status. The controller status is generated from a combination of software conditions and the status of its components. For example, the status **ok** is reported for properly functioning controllers. Controllers contain 2-port Fibre Channel interface cards, 2-port Ethernet I/O interface cards, or a mix of both types.

The components in a FA-400 controller include:

- **Ethernet Port (CTm.ETHn):**

Ethernet ports report status and link speed. Each controller contains at least 1x1GbE port for management (including one used for remote array administration) and 2x10GbE ports for replication.

- **FAN (CTm.FANn):**

Each controller contains multiple cooling fans that report status. Fan replacement is a service technician operation.

- **Fibre Channel Port (CTm.FCn):**

Fibre Channel ports report status and link speed. Most Fibre Channel-based controllers contain redundant 2-port Fibre Channel host bus adapters. Certain controllers, such as the FA-405 controller and controllers with mixed I/O configurations, support a single Fibre Channel adapter. To see the Fibre Channel port details for each FlashArray model, refer to the Products page at <https://support.purestorage.com>.

- **InfiniBand Port (CTm.IBn):**

Controllers within an FA-400 series array communicate via redundant InfiniBand interfaces located on a single PCI Express host bus adapter. InfiniBand ports report status and communication speed.

- **Power Supply (CTm.PWRn):**

Each controller contains redundant power supplies that report status. Power supply replacement is a service technician operation.

- **SAS Ports (CTm.SASn):**

Multiple SAS ports handle communication between the array's controllers and the shelf's flash modules and NVRAM modules. **Note:** The FA-405 model supports only a single SAS card.

- **Temperature Sensor (CTm.TMPn):**

Each controller chassis contains multiple temperature sensors that report their status and the ambient temperature at various points within the chassis. In steady-state operation, Purity//FA generates alerts when a temperature sensor reports a value outside of normal operating range.

FA-400 Storage Shelf Components (SHm)

Purity//FA reports an overall storage shelf (**SHm**) status generated from a combination of software conditions and the status of its components.

The components in a FA-400 storage shelf include:

- **Shelf (SHm):**

The first two storage shelves contain both storage bays and NVRAM bays. Additional shelves, if present, contain only storage bays.

- **Storage Bay (SHm.BAYn):**

Each storage shelf contains a row of hot-swappable storage bays, which contain flash modules used for the array's data. Storage bays are indexed from left to right starting at 0.

- **Fan (SHm.FANn):**

Each of a shelf's power and cooling modules (PCMs) reports cooling fan status.

- **IOM (SHm.IOMn):**

Redundant I/O Modules (IOMs) accessed from the rear of the shelf chassis contain the shelf's SAS ports.

- **Power Supply (SHm.PWRn):**

The power supply sub-component within each of a shelf's power and cooling modules (PCMs) reports its operating status. Power supply replacement is a service technician operation.

- **SAS Port (SHm.SASn):**

Multiple SAS ports handle communication between the array's controllers and the shelf's flash modules and NVRAM modules. **Note:** The FA-405 model supports only a single SAS card.

- **Temperature Sensor (SHm.TMPn):**

Each controller chassis contains multiple temperature sensors that report their status and the ambient temperature at various points within the chassis. In steady-state operation, Purity//FA generates alerts when a temperature sensor reports a value outside of normal operating range.

Viewing Hardware Component Details

The **purehw list** command displays information about array hardware components that are capable of reporting their status. The display is primarily useful for diagnosing hardware-related problems.

If one or more component names are specified in the command line, information is displayed for those components only. If component names are not specified in the command line, information is displayed for all components.

Include the **--spec** option to display the hardware specifications for each component wherever applicable. Hardware specifications, all of which are unique to Pure Storage, include FlashArray hardware model names, part numbers, and serial numbers. Component part numbers and serial numbers are also printed on the physical hardware.

Include the **--type** option to display information for all components of the specified type.

The **purehw list** output includes the following columns:

Status

Each component reports its status as either:

ok

Functioning properly at full capacity.

critical

Not functioning or requiring immediate attention.

degraded

Functioning, but not at full capability due to a non-fatal failure.

device_off

Installed, but powered off.

identifying

Functioning, but is not yet initialized.

not_installed

Does not appear to be installed.

unknown

Insufficient information to determine a state for this device.

Identify

State (on or off) of an LED used to visually identify the component. (Relevant only for controllers, NVRAM bays, storage bays, and storage shelves.)

Slot

Slot number occupied by the PCI Express card that hosts the component. (Relevant only for ports hosted by PCI Express cards in controller chassis.)

Index

Number that identifies the relative position of a hardware component within the array.

Speed

Speed at which the component is operating. (Relevant only for interface ports [bits/second].)

Temperature

Temperature reported by the component. (Relevant only for temperature sensors.)

Voltage

Input voltage (VIN) of the power supplies in the chassis. Only applies to //M models.

Changing Hardware Component Attributes

FlashArray controllers, storage shelves, NVRAM bays, and storage bays are equipped with LEDs that are illuminated to identify a physical component. The **purehw setattr --identify** command turns a component's identifying light on or off. The Identify column in the **purehw list** output displays the current state of the identifying light for each component that has one.

Storage shelf control panels contain numeric displays in which numbers can be set to uniquely identify shelves in multi-shelf arrays. Run the **purehw setattr --index** command to set the top-level storage shelf number. Once the top-level storage shelf number has been set, the number becomes part of the name of each component in the shelf. For example, the `purehw setattr --index 2 SH0` command sets shelf `SH0` to `SH2`, and all of the components in the shelf automatically assume names in the form `SH2.COMPONENT-NAME` (e.g., `SH2.BAY0`, `SH2.FAN1`, etc.).

Examples

Example 1

```
purehw list SH0.BAY11 SH0.BAY12
```

Displays information for flash modules in slots `11` and `12` of storage shelf `0`.

Example 2

```
purehw list --type fan
```

Displays information for all cooling fans in the array.

Example 3

```
purehw setattr --identify on CH0.BAY1
```

Illuminates the identifying LED for storage bay 1 in shelf CH0.

Example 4

```
purehw setattr --index 2 SH0
```

Sets shelf SH0 and all of its components to SH2. For example, sets SH0 to SH2, SH0.BAY0 to SH2.BAY0, SH0.FAN1 to SH2.FAN1, and so on).

See Also

[puredrive\(1\)](#)

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